Matrix Algebra

Goals

Study matrix algebra with introduction to some applications

1. Algebra of matrix: addition, scalar multiplication, transposition, product
2. Matrix multiplication
   - "Matrix transformation" such as rotation are important tool in geometry and in return geometry provides a "pictures" of the matrices
   - Matrix multiplication is related to directed graphs
3. Matrix inverse
4. Introduction to matrix LU-factorization

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5. LU Factorization

Matrix

A rectangular array of numbers is called a matrix, denoted by uppercase letters: A, B, C...

$$\underbrace{\begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix}}_{n \text{ columns}} \left.\vphantom{\begin{bmatrix} a \\ a \\ a \\ \vdots \\ a \end{bmatrix}}\right\} m \text{ rows}$$

- an $m \times n$ matrix or **size** $m \times n$.
- Simple notation $A = [a_{ij}]$
- (i, j) - **entry** : element lies in row i and column j, denoted by a_{ij}

Row Matrix or Row Vector

A matrix of size $1 \times n$ is called a row matrix

$$\begin{bmatrix} 1 & 2 & 4 & 9 \end{bmatrix}$$

Column Matrix or Column Vector

A matrix of size $m \times 1$ is called a column matrix

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

Square matrix

A $m \times m$ matrix is a square matrix

$$\begin{bmatrix} 1 & 2 \\ 4 & 9 \end{bmatrix}$$

Example

$$B = \begin{bmatrix} 1 & 2 & -1 \\ 0 & 5 & 6 \end{bmatrix}$$

is a 2×3 matrix.
The $(2, 1)$ - entry is

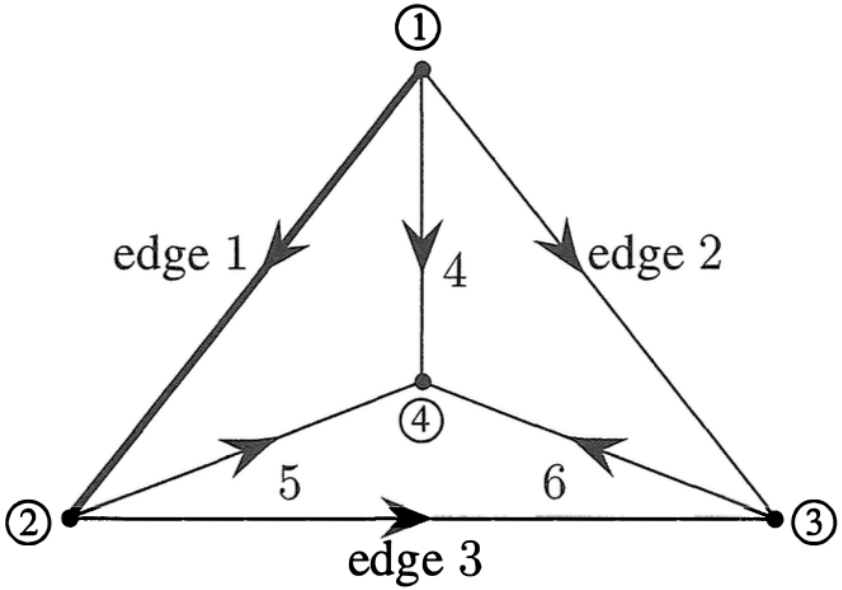
$$b_{21} = 0$$

Matrix and dataset

		Monthly sales for goods		
		G1	G2	G3
Sold to	C1	7	3	4
customer	C2	1	5	6

Graph and Incidence Matrix

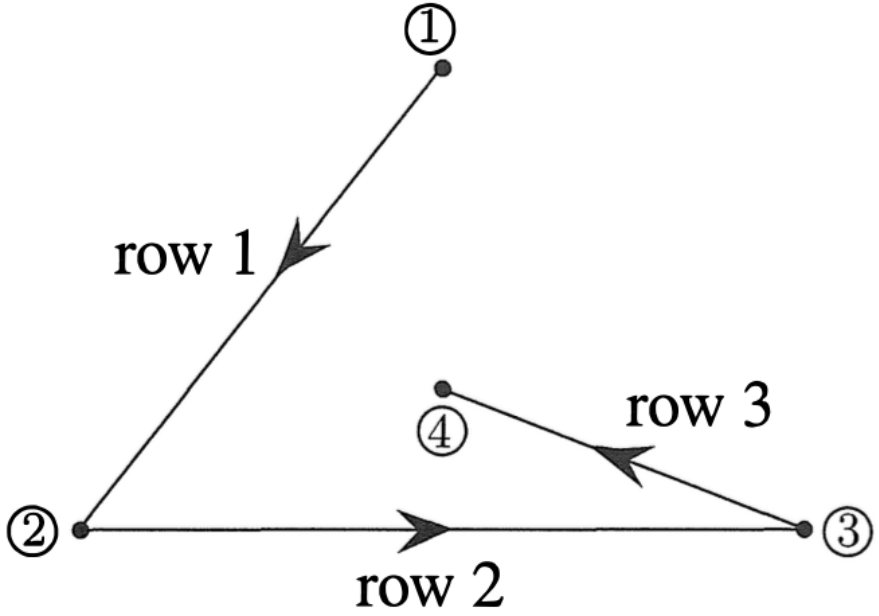
Graph is a model consisting of nodes connected by edges. An incidence matrix tell hows the nodes are connected by the edges. Every entry of an incidence matrix is 0, -1 or 1.



$$
\begin{array}{cc}
 & \textbf{node} \\
 & \begin{array}{cccc} 1 & 2 & 3 & 4 \end{array} \\
A = & \begin{bmatrix} -1 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & -1 & 1 & 0 \\ -1 & 0 & 0 & 1 \\ 0 & -1 & 0 & 1 \\ 0 & 0 & -1 & 1 \end{bmatrix} \begin{array}{c} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \end{array} \quad \textbf{edge}
\end{array}
$$

In the first row -1, 1, 0, 0 show that the first edge goes from node 1 to node 2 (-1 for node 1 because the arrow goes out, +1 for node 2 with arrow in)

This graph is **complete** - *every pair of nodes is connected by an edge.*

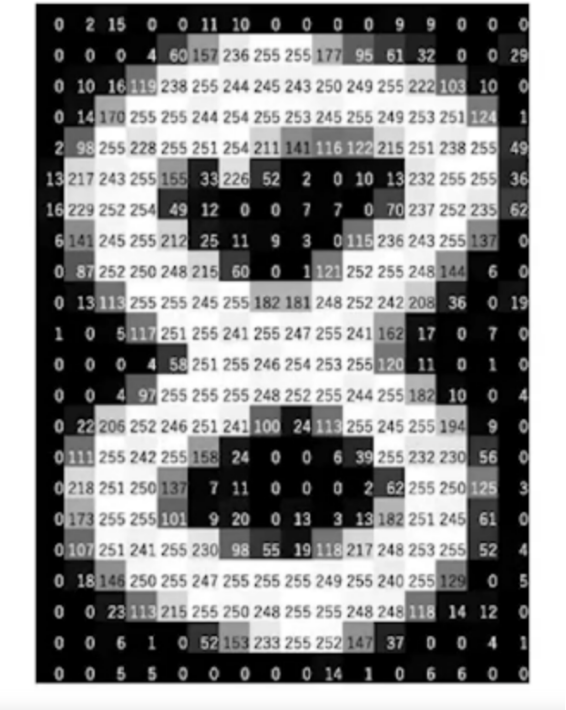


$$B = \begin{array}{c} \begin{array}{cccc} ① & ② & ③ & ④ \end{array} \\ \begin{bmatrix} 1 & -1 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -1 \end{bmatrix} \end{array} \begin{array}{c} 1 \\ 2 \\ 3 \end{array} \quad \text{edge}$$

This graph is a **tree** - a graph has **no closed loops**

B&W or Greyscale Image

Image is distorted into small boxes called **pixels**. Dimension of the image is simply the number of pixels across the image's height(x) and width(y), e.g, 24×16

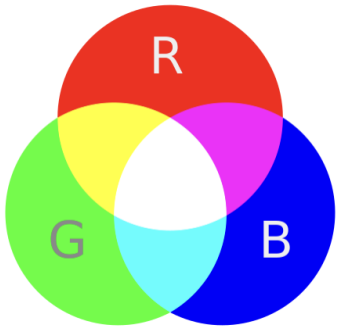


0	2	15	0	0	11	10	0	0	0	0	9	9	0	0	0
0	0	0	4	60	157	236	255	255	177	95	61	32	0	0	29
0	10	16	119	238	255	244	245	243	250	249	255	222	103	10	0
0	14	170	255	255	244	254	255	253	245	255	249	253	251	124	1
2	98	255	228	255	251	254	211	141	116	122	215	251	238	255	49
13	217	243	255	155	33	226	52	2	0	10	13	232	255	255	36
16	229	252	254	49	12	0	0	7	7	0	70	237	252	235	62
6	141	245	255	212	25	11	9	3	0	115	236	243	255	137	0
0	87	252	250	248	215	60	0	1	121	252	255	248	144	6	0
0	13	113	255	255	245	255	182	181	248	252	242	208	36	0	19
1	0	5	117	251	255	241	255	247	255	241	162	17	0	7	0
0	0	0	4	58	251	255	246	254	253	255	120	11	0	1	0
0	0	4	97	255	255	255	248	252	255	244	255	182	10	0	4
0	22	206	252	246	251	241	100	24	113	255	245	255	194	9	0
0	111	255	242	255	158	24	0	0	6	39	255	232	230	56	0
0	218	251	250	137	7	11	0	0	0	2	62	255	250	125	3
0	173	255	255	101	9	20	0	13	3	13	182	251	245	61	0
0	107	251	241	255	230	98	55	19	118	217	248	253	255	52	4
0	18	146	250	255	247	255	255	255	249	255	240	255	129	0	5
0	0	23	113	215	255	250	248	255	255	248	248	118	14	12	0
0	0	6	1	0	52	153	233	255	252	147	37	0	0	4	1
0	0	5	5	0	0	0	0	0	14	1	0	6	6	0	0

Each of these pixels is denoted as a numerical value, called **Pixel Values** which denote the intensity of the pixels. For a greyscale or b&w image, pixel values range from 0 to 255.

RGB Color Images

The RGB color model is an additive color model in which the red, green and blue primary colors of light are added together in various ways to reproduce a broad array of colors.



RGB image can be viewed as three different images (a red scale image, a green scale image and a blue scale image)

Full color image along with its R, G, and B

Equal matrices

Two matrices A and B are equal, denoted by

$$A = B$$

if

1. They have the same size.
2. Corresponding entries are equal.

Example

Given

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, B = \begin{bmatrix} 1 & 2 & -1 \\ 3 & 0 & 1 \end{bmatrix} \text{ and } C = \begin{bmatrix} 1 & 0 \\ -2 & 2 \end{bmatrix}$$

Discuss the possibility that $A = B$, $B = C$, $A = C$.

Example

Given

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \; B = \begin{bmatrix} 1 & 2 & -1 \\ 3 & 0 & 1 \end{bmatrix} \text{ and } C = \begin{bmatrix} 1 & 0 \\ -2 & 2 \end{bmatrix}$$

Discuss the possibility that $A = B$, $B = C$, $A = C$.

Solution

- *$A = B$ is impossible because A and B are of different sizes: A is 2×2 whereas B is 2×3.*
- *Similarly, $B = C$ is impossible.*
- *$A = C$ is possible provided that corresponding entries are equal:*

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -2 & 2 \end{bmatrix}$$

means that $a = 1$, $b = 0$, $c = -2$ and $d = 2$

Matrix addition

If A and B are matrices of the same size, their sum $A + B$ is the matrix formed by **adding corresponding entries**.

If $A = [a_{ij}]$ and $B = [b_{ij}]$ then

$$A + B = [a_{ij} + b_{ij}]$$

in other words

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} + \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1n} \\ b_{21} & b_{22} & \cdots & b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ b_{n1} & b_{m2} & \cdots & b_{mn} \end{bmatrix}$$

$$= \begin{bmatrix} a_{11} + b_{11} & a_{12} + b_{12} & \cdots & a_{1n} + b_{1n} \\ a_{21} + b_{21} & a_{22} + b_{22} & \cdots & a_{2n} + b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} + b_{n1} & a_{m2} + b_{m2} & \cdots & a_{mn} + b_{mn} \end{bmatrix}$$

Example

Sales data in a fast food restaurant chain in two days are given by

$$D1 = \begin{array}{r} \\ South\ stand \\ North\ stand \\ East\ stand \end{array} \begin{array}{c} \begin{array}{ccc} Peanuts & Hotdogs & Soda \end{array} \\ \begin{pmatrix} 120 & 250 & 305 \\ 207 & 140 & 419 \\ 29 & 120 & 190 \end{pmatrix} \end{array}$$

$$D2 = \begin{array}{r} \\ South\ stand \\ North\ stand \\ East\ stand \end{array} \begin{array}{c} \begin{array}{ccc} Peanuts & Hotdogs & Soda \end{array} \\ \begin{pmatrix} 110 & 230 & 280 \\ 300 & 155 & 389 \\ 35 & 117 & 201 \end{pmatrix} \end{array}$$

then

$$D1 + D2 = \begin{array}{r} \\ South\ stand \\ North\ stand \\ East\ stand \end{array} \begin{array}{c} \begin{array}{ccc} Peanuts & Hotdogs & Soda \end{array} \\ \begin{pmatrix} 120+110 & 250+230 & 305+305 \\ 207+300 & 140+155 & 419+389 \\ 29+35 & 120+117 & 190+201 \end{pmatrix} \end{array}$$

represents total sales in 2 days of each product in each branch

Example

Find a, b, c such that

$$\begin{bmatrix} a & b & c \end{bmatrix} + \begin{bmatrix} c & a & b \end{bmatrix} = \begin{bmatrix} 3 & 2 & -1 \end{bmatrix}$$

Example

Find a, b, c such that

$$\begin{bmatrix} a & b & c \end{bmatrix} + \begin{bmatrix} c & a & b \end{bmatrix} = \begin{bmatrix} 3 & 2 & -1 \end{bmatrix}$$

Solution

Add the matrices on the left side to obtain

$$\begin{bmatrix} a+c & b+a & c+b \end{bmatrix} = \begin{bmatrix} 3 & 2 & -1 \end{bmatrix}$$

This gives $a + c = 3$, $b + a = 2$ *and* $c + b = -1$
Solving these yields $a = 3, b = -1, c = 0$.

Zero matrix

The $m \times n$ matrix in which every entry is zero is called the $m \times n$ zero matrix and is denoted as $\mathbf{0}$ (or $\mathbf{0}_{mn}$)

$$\mathbf{0}_{23} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Negative matrix

Negative of matrix A, denoted by $-A$ obtained by multiplying each entry of A by -1

$$-A = [-a_{ij}]$$

Difference

If A and B are two $m \times n$ matrices, their difference $A - B$ is defined by

$$A - B = A + (-B)$$

If $A = [a_{ij}]$ and $B = [b_{ij}]$ then $A - B = [a_{ij} - b_{ij}]$

Example

Sales data in a fast food restaurant chain in two days are given by

$$D1 = \begin{array}{r} \\ South\ stand \\ North\ stand \\ East\ stand \end{array} \begin{array}{c} \begin{array}{ccc} Peanuts & Hotdogs & Soda \end{array} \\ \begin{pmatrix} 120 & 250 & 305 \\ 207 & 140 & 419 \\ 29 & 120 & 190 \end{pmatrix} \end{array}$$

$$D2 = \begin{array}{r} \\ South\ stand \\ North\ stand \\ East\ stand \end{array} \begin{array}{c} \begin{array}{ccc} Peanuts & Hotdogs & Soda \end{array} \\ \begin{pmatrix} 110 & 230 & 280 \\ 300 & 155 & 389 \\ 35 & 117 & 201 \end{pmatrix} \end{array}$$

then

$$D1 - D2 = \begin{array}{r} \\ South\ stand \\ North\ stand \\ East\ stand \end{array} \begin{array}{c} \begin{array}{ccc} Peanuts & Hotdogs & Soda \end{array} \\ \begin{pmatrix} 120-110 & 250-230 & 305-280 \\ 207-300 & 140-155 & 419-389 \\ 29-35 & 120-117 & 190-201 \end{pmatrix} \end{array}$$

represents the difference of sales of each product in each branch

Example

Let

$$A = \begin{bmatrix} 3 & -1 & 0 \\ 1 & 2 & -4 \end{bmatrix}, B = \begin{bmatrix} 1 & -1 & 1 \\ -2 & 0 & 6 \end{bmatrix}, C = \begin{bmatrix} 1 & 0 & -2 \\ 3 & 1 & 1 \end{bmatrix}$$

Compute $-A$, $A - B$, $A + B - C$.

Example

Let

$$A = \begin{bmatrix} 3 & -1 & 0 \\ 1 & 2 & -4 \end{bmatrix}, B = \begin{bmatrix} 1 & -1 & 1 \\ -2 & 0 & 6 \end{bmatrix}, C = \begin{bmatrix} 1 & 0 & -2 \\ 3 & 1 & 1 \end{bmatrix}$$

Compute $-A$, $A - B$, $A + B - C$.

Solution

$$-A = \begin{bmatrix} -3 & 1 & 0 \\ -1 & -2 & 4 \end{bmatrix}$$

$$A - B = \begin{bmatrix} 3-1 & -1-(-1) & 0-1 \\ 1-(-2) & 2-0 & -4-6 \end{bmatrix} = \begin{bmatrix} 2 & 0 & -1 \\ 3 & 2 & -10 \end{bmatrix}$$

$$A + B - C = \begin{bmatrix} 3+1-1 & -1-1-0 & 0+1-(-2) \\ 1-2-3 & 2+0-1 & -4+6-1 \end{bmatrix} = \begin{bmatrix} 3 & -2 & 3 \\ -4 & 1 & 1 \end{bmatrix}$$

Example

Solve

$$\begin{bmatrix} 3 & 2 \\ -1 & 1 \end{bmatrix} + X = \begin{bmatrix} 1 & 0 \\ -1 & 2 \end{bmatrix}$$

where X is a matrix.

Example

Solve

$$\begin{bmatrix} 3 & 2 \\ -1 & 1 \end{bmatrix} + X = \begin{bmatrix} 1 & 0 \\ -1 & 2 \end{bmatrix}$$

where X is a matrix.

Solution

Substract the matrix $\begin{bmatrix} 3 & 2 \\ -1 & 1 \end{bmatrix}$ *from both side*

$$X = \begin{bmatrix} 1 & 0 \\ -1 & 2 \end{bmatrix} - \begin{bmatrix} 3 & 2 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} -2 & -2 \\ 0 & 1 \end{bmatrix}$$

Scalar multiplication

Definition

If $A = [a_{ij}]$ is any matrix and k is any number, the scalar multiple kA is the matrix obtained from A by multiplying each entry of A by k

$$kA = [ka_{ij}]$$

Example

$$D = \begin{array}{r} \\ South\ stand \\ North\ stand \\ East\ stand \end{array} \begin{array}{c} \begin{array}{ccc} Peanuts & Hotdogs & Soda \end{array} \\ \begin{pmatrix} 120 & 250 & 305 \\ 207 & 140 & 419 \\ 29 & 120 & 190 \end{pmatrix} \end{array}$$

represent sales data in a day then

$$7D = \begin{array}{r} \\ South\ stand \\ North\ stand \\ East\ stand \end{array} \begin{array}{c} \begin{array}{ccc} Peanuts & Hotdogs & Soda \end{array} \\ \begin{pmatrix} 7*120 & 7*250 & 7*305 \\ 7*207 & 7*140 & 7*419 \\ 7*29 & 7*120 & 7*190 \end{pmatrix} \end{array}$$

refers to the sales data in a week

Example

Let

$$A = \begin{bmatrix} 3 & -1 & 4 \\ 2 & 0 & 1 \end{bmatrix}, B = \begin{bmatrix} 1 & 2 & -1 \\ 0 & 3 & 2 \end{bmatrix}.$$

Compute $5A$, $\frac{1}{2}B$, $3A - 2B$.

Solution

$$5A = \begin{bmatrix} 15 & -5 & 20 \\ 10 & 0 & 5 \end{bmatrix}, \quad \frac{1}{2}B = \begin{bmatrix} 0.5 & 1 & -0.5 \\ 0 & 1.5 & 1 \end{bmatrix}$$

$$3A - 2B = \begin{bmatrix} 9 & -3 & 12 \\ 6 & 0 & 3 \end{bmatrix} - \begin{bmatrix} 2 & 4 & -2 \\ 0 & 6 & 4 \end{bmatrix} = \begin{bmatrix} 7 & -7 & 14 \\ 6 & -6 & -1 \end{bmatrix}$$

Properties of matrix addition and scalar multiplication

Let A, B, and C denote arbitrary $m \times n$ matrices where m and n are fixed. Let k and p denote arbitrary real numbers. Then

1. $A + B = B + A$ (commutative law).
2. $A + (B + C) = (A + B) + C$ (associative law)
3. $0 + A = A$ for each A.
4. $A + (-A) = 0$
5. $k(A + B) = kA + kB$.
6. $(k + p)A = kA + pA$.
7. $(kp)A = k(pA)$.
8. $1A = A$.

Example

Simplify $2(A+3C)-3(2C-B)-3[2(2A+B-4C)-4(A-2C)]$ where A, B, and C are all matrices of the same size.

Example

Simplify $2(A+3C)-3(2C-B)-3[2(2A+B-4C)-4(A-2C)]$ where A, B, and C are all matrices of the same size.

Solution

$$
\begin{aligned}
&2(A+3C)-3(2C-B)-3[2(2A+B-4C)-4(A-2C)] \\
&= 2A+6C-6C+3B-3[4A+2B-8C-4A+8C] \\
&= 2A+3B-3(2B) \\
&= 2A+3B-6B \\
&= 2A-3B
\end{aligned}
$$

Transpose of a Matrix

If A is an $m \times n$ matrix, the transpose of A, written A^T , is the $n \times m$ matrix whose rows are just the columns of A in the same order.

$$a_{ij}^T = a_{ji}$$

The first row of A^T is the first column of A (that is it consists of the entries of column 1 in order). Similarly the second row of A^T is the second column of A, and so on.

Example

Write down the transpose of each of the following matrices.

$$A = \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix}, B = \begin{bmatrix} 5 & 2 & 6 \end{bmatrix}, C = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 4 & 6 \end{bmatrix}, D = \begin{bmatrix} 3 & 1 & -1 \\ 1 & 3 & 2 \\ -1 & 2 & 1 \end{bmatrix}$$

Example

Write down the transpose of each of the following matrices.

$$A = \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix}, B = \begin{bmatrix} 5 & 2 & 6 \end{bmatrix}, C = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 4 & 6 \end{bmatrix}, D = \begin{bmatrix} 3 & 1 & -1 \\ 1 & 3 & 2 \\ -1 & 2 & 1 \end{bmatrix}$$

Solution

$$A^T = \begin{bmatrix} 1 & 3 & 2 \end{bmatrix}, B^T = \begin{bmatrix} 5 \\ 2 \\ 6 \end{bmatrix}, C^T = \begin{bmatrix} 1 & 3 & 4 \\ 2 & 4 & 6 \end{bmatrix}, D^T = \begin{bmatrix} 3 & 1 & -1 \\ 1 & 3 & 2 \\ -1 & 2 & 1 \end{bmatrix} = D$$

Another way to think of transposition

Main diagonal consists of the elements a_{11}, a_{22} , . . .

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} a_{11} \\ a_{21} \end{bmatrix}$$

Forming the transpose of a matrix A can be viewed as "flipping" A about its main diagonal, or as "rotating" A through 180° about the line containing the main diagonal

Special types of matrix

Definition (Symmetric)

A matrix A is symmetric if $A^T = A$

Definition (Skew symmetric)

A matrix A is skew symmetric if $A^T = -A$

Example

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix} \text{ is a symmetric matrix}$$

$$B = \begin{bmatrix} 0 & 2 & 3 \\ -2 & 0 & -4 \\ -3 & 4 & 0 \end{bmatrix} \text{ is a skew symmetric matrix}$$

Properties of Matrix Transpose

Let A and B denote matrices of the same size, and let k denote a scalar.

1. If A is an $m \times n$ matrix, then A^T is an $n \times m$ matrix.
2. $(A^T)^T = A$.
3. $(kA)^T = kA^T$.
4. $(A + B)^T = A^T + B^T$.

Example

Solve for A if

$$\left(2A^T - 3\begin{bmatrix} 1 & 2 \\ -1 & 1 \end{bmatrix}\right)^T = \begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix}$$

Example

Solve for A if

$$\left(2A^T - 3\begin{bmatrix} 1 & 2 \\ -1 & 1 \end{bmatrix}\right)^T = \begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix}$$

Solution

Applying the above theorem for the left hand side, we have

$$LHS = 2(A^T)^T - 3\begin{bmatrix} 1 & 2 \\ -1 & 1 \end{bmatrix}^T = 2A - 3\begin{bmatrix} 1 & -1 \\ 2 & 1 \end{bmatrix}$$

The equation becomes

$$2A - 3\begin{bmatrix} 1 & -1 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix} \Rightarrow 2A = \begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix} + 3\begin{bmatrix} 1 & -1 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 5 & 0 \\ 5 & 5 \end{bmatrix}$$

So $A = \frac{1}{2}\begin{bmatrix} 5 & 0 \\ 5 & 5 \end{bmatrix} = \begin{bmatrix} 2.5 & 0 \\ 2.5 & 2.5 \end{bmatrix}$

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Vector Form of Linear System

The linear system

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2 \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m \end{cases}$$

with coefficient matrix A and constant matrix **b** can be written in **vector form**

$$\begin{bmatrix} a_{11} \\ a_{21} \\ \cdots \\ a_{n1} \end{bmatrix} x_1 + \begin{bmatrix} a_{21} \\ a_{22} \\ \cdots \\ a_{n2} \end{bmatrix} x_2 + \cdots + \begin{bmatrix} a_{1n} \\ a_{2n} \\ \cdots \\ a_{nn} \end{bmatrix} x_n \equiv \begin{bmatrix} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n \\ \cdots \\ a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n \end{bmatrix} = \mathbf{b}$$

in other words

$$\mathbf{col}_1(A)x_1 + \mathbf{col}_2(A)x_2 + \cdots + \mathbf{col}_n(A)x_n = \mathbf{b}$$

Example

Consider the linear system $A\mathbf{x} = \mathbf{b}$ where

$$A = \begin{bmatrix} 3 & 1 & 2 \\ 4 & -5 & 6 \\ 0 & 7 & -3 \\ -1 & 2 & 0 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 4 \\ 1 \\ 0 \\ 2 \end{bmatrix}$$

then we can represent the system in terms of equation of columns

$$x_1 \begin{bmatrix} 3 \\ 4 \\ 0 \\ -1 \end{bmatrix} + x_2 \begin{bmatrix} 1 \\ -5 \\ 7 \\ 2 \end{bmatrix} + x_3 \begin{bmatrix} 2 \\ 6 \\ -3 \\ 0 \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \\ 0 \\ 2 \end{bmatrix}$$

A consequence of vector form of a linear system is that we can find a condition to guarantee the existence of solution to a linear system

Theorem

A linear system with coefficient matrix A and constant matrix $\boldsymbol{b}$ has at least one solution (that is the system is consistent) if and only if $\boldsymbol{b}$ can be expressed as a linear combination of the columns of matrix A.

Being motivated by vector form of linear system and representing variables in a column matrix (also called a vector), we define product of a matrix and a vector in a same way

Matrix-vector product

Let A be an $m \times n$ matrix. Write

$$A = \begin{bmatrix} \mathbf{a}_1 & \dots \mathbf{a}_n \end{bmatrix}$$

where $\mathbf{a}_j$ are the columns of A and

$$\mathbf{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$$

then

$$A\mathbf{x} = \mathbf{a}_1 x_1 + \cdots + \mathbf{a}_n x_n$$

Example

Let

$$A = \begin{bmatrix} 2 & -1 & -3 \\ 4 & 2 & -2 \end{bmatrix} \text{ and } \mathbf{x} = \begin{bmatrix} 2 \\ -3 \\ 4 \end{bmatrix}$$

then

$$\begin{aligned} A\mathbf{x} &= 2\mathbf{col}_1(A) - 3\mathbf{col}_2(A) + 4\mathbf{col}_3(A) \\ &= 2\begin{bmatrix} 2 \\ 4 \end{bmatrix} - 3\begin{bmatrix} -1 \\ 2 \end{bmatrix} + 4\begin{bmatrix} -3 \\ -2 \end{bmatrix} = \begin{bmatrix} -5 \\ -6 \end{bmatrix} \end{aligned}$$

Example - Matrix Form of Linear System

The system of linear equation

$$\begin{cases} -2x \qquad\quad + \quad z = 5 \\ \quad 2x + 3y - 4z = 7 \\ \quad 3x + 2y + 2z = 3 \end{cases}$$

can be written is matrix form

$$A\mathbf{x} = \mathbf{b}$$

with $A = \begin{bmatrix} -2 & 0 & 1 \\ 2 & 3 & -4 \\ 3 & 2 & 2 \end{bmatrix}$, $\mathbf{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 5 \\ 7 \\ 3 \end{bmatrix}$

Exercise

Determine the matrix A, $\mathbf{x}$, $\mathbf{b}$ to write the following system of linear equations in the form $A\mathbf{x} = \mathbf{b}$.

$$\begin{cases} 5x_1 - x_2 + 2x_3 + x_4 - 3x_5 = 8 \\ x_1 + x_2 + 3x_3 - 5x_4 + 2x_5 = -2 \\ -x_1 + x_2 - 2x_3 - 3x_5 = 0 \end{cases}$$

Matrix multiplication or matrix product

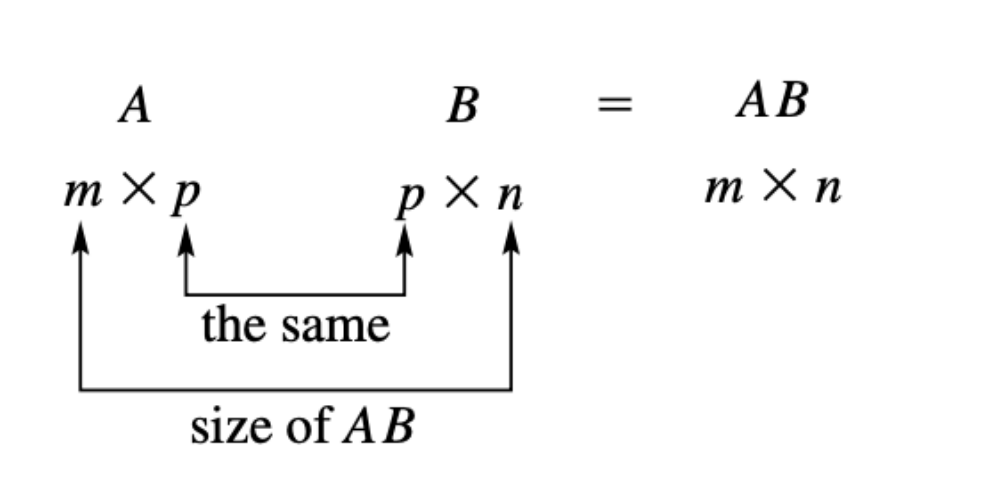
Definition

Let A be an $m \times p$ matrix and B be an $p \times n$ matrix. Write $B = \begin{bmatrix} \mathbf{b}_1 & \dots & \mathbf{b}_n \end{bmatrix}$ where $\mathbf{b}_j$ is column j of B then the product matrix AB is the $m \times n$ matrix defined by

$$AB = A \begin{bmatrix} \mathbf{b}_1 & \dots & \mathbf{b}_n \end{bmatrix} = \begin{bmatrix} A\mathbf{b}_1 & \dots & A\mathbf{b}_n \end{bmatrix}$$

Column j of AB this the matrix-vector product of A and column j of B.

Compatible rule



Example

Compute AB if $A = \begin{bmatrix} 2 & 3 & 5 \\ 1 & 4 & 7 \\ 0 & 1 & 8 \end{bmatrix}$ and $B = \begin{bmatrix} 8 & 9 \\ 7 & 2 \\ 6 & 1 \end{bmatrix}$

Example

Compute AB if $A = \begin{bmatrix} 2 & 3 & 5 \\ 1 & 4 & 7 \\ 0 & 1 & 8 \end{bmatrix}$ and $B = \begin{bmatrix} 8 & 9 \\ 7 & 2 \\ 6 & 1 \end{bmatrix}$

Solution

Columns of B *are* $\boldsymbol{b}_1 = \begin{bmatrix} 8 \\ 7 \\ 6 \end{bmatrix}$ *and* $\boldsymbol{b}_2 = \begin{bmatrix} 9 \\ 2 \\ 1 \end{bmatrix}$. *So*

$$A\boldsymbol{b}_1 = \begin{bmatrix} 2 & 3 & 5 \\ 1 & 4 & 7 \\ 0 & 1 & 8 \end{bmatrix} \begin{bmatrix} 8 \\ 7 \\ 6 \end{bmatrix} = \begin{bmatrix} 67 \\ 78 \\ 55 \end{bmatrix}, \quad A\boldsymbol{b}_1 = \begin{bmatrix} 2 & 3 & 5 \\ 1 & 4 & 7 \\ 0 & 1 & 8 \end{bmatrix} \begin{bmatrix} 9 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 29 \\ 24 \\ 10 \end{bmatrix}$$

Thus $AB = \begin{bmatrix} A\boldsymbol{b}_1 & A\boldsymbol{b}_2 \end{bmatrix} = \begin{bmatrix} 67 & 29 \\ 78 & 24 \\ 55 & 10 \end{bmatrix}$

Practice Problem

Write down the order of the matrices

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \\ 3 & 1 \end{bmatrix} \qquad B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

Find (if possible) AB and BA

(i, j)-entry of matrix product

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1p} \\ a_{21} & a_{22} & \dots & a_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mp} \end{bmatrix}, \quad B = \begin{bmatrix} b_{11} & b_{12} & \dots & b_{1n} \\ b_{21} & b_{22} & \dots & b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ b_{p1} & b_{p2} & \dots & b_{pn} \end{bmatrix}$$

The j column of AB is of form

$$A\mathbf{col}_j(B) = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1p} \\ a_{21} & a_{22} & \dots & a_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mp} \end{bmatrix} \begin{bmatrix} b_{1j} \\ b_{2j} \\ \vdots \\ b_{pj} \end{bmatrix} = b_{1j}\mathbf{col}_1(A) + \cdots + b_{pj}\mathbf{col}_p(A)$$

$$A\mathbf{col}_j(B) = \begin{bmatrix} a_{11}b_{1j} \\ a_{21}b_{1j} \\ \vdots \\ a_{m1}b_{1j} \end{bmatrix} + \cdots + \begin{bmatrix} a_{1p}b_{pj} \\ a_{2p}b_{pj} \\ \vdots \\ a_{mp}b_{pj} \end{bmatrix}$$

$$= \begin{bmatrix} a_{11}b_{1j} + a_{12}b_{2j} + \cdots + a_{1p}b_{pj} \\ a_{21}b_{1j} + a_{22}b_{2j} + \cdots + a_{2p}b_{pj} \\ \vdots \\ a_{m1}b_{1j} + a_{m2}b_{2j} + \cdots + a_{mp}b_{pj} \end{bmatrix} = \begin{bmatrix} \sum_{k=1}^{p} a_{1k}b_{kj} \\ \sum_{k=1}^{p} a_{2k}b_{kj} \\ \vdots \\ \sum_{k=1}^{p} a_{mk}b_{kj} \end{bmatrix}$$

So the (i, j) entry of AB is

$$a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{ip}b_{pj} = \sum_{k=1}^{p} a_{ik}b_{kj}$$

Rule to multiply matrices

If A is an $m \times n$ matrix and B is an $n \times q$ matrix then

$$C = AB$$

is an $m \times q$ matrix and c_{ij} **is found by multiplying the ith row of A by the jth column of B.**

$$\begin{array}{c} \\ \text{row}_i\,(A) \end{array} \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1p} \\ a_{21} & a_{22} & \cdots & a_{2p} \\ \vdots & \vdots & & \vdots \\ a_{i1} & a_{i2} & \cdots & a_{ip} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mp} \end{bmatrix} \begin{array}{c} \text{col}_j\,(B) \\ \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1j} & \cdots & b_{1n} \\ b_{21} & b_{22} & \cdots & b_{2j} & \cdots & b_{2n} \\ \vdots & \vdots & & \vdots & & \vdots \\ b_{p1} & b_{p2} & \cdots & b_{pj} & \cdots & b_{pn} \end{bmatrix} \end{array}$$

$$= \begin{bmatrix} c_{11} & c_{12} & \cdots & c_{1n} \\ c_{21} & c_{22} & \cdots & c_{2n} \\ \vdots & \vdots & c_{ij} & \vdots \\ c_{m1} & c_{m2} & \cdots & c_{mn} \end{bmatrix}$$

$$(\text{row}_i\,(A))^T \cdot \text{col}_j\,(B) = \sum_{k=1}^{p} a_{ik} b_{kj} = c_{ij}$$

Example

Compute product of two matrices of size 2×3 and 3×4 respectively. The product is of size 2×4. So we denote the entries of the product by c_{ij} with row index $i = 1, 2$ and column index $j = 1, 2, 3, 4$

$$\begin{bmatrix} 2 & 1 & 0 \\ 1 & 0 & 4 \end{bmatrix} \begin{bmatrix} 3 & 1 & 2 & 1 \\ 1 & 0 & 1 & 2 \\ 5 & 4 & 1 & 1 \end{bmatrix} = \begin{bmatrix} c_{11} & c_{12} & c_{13} & c_{14} \\ c_{21} & c_{22} & c_{23} & c_{24} \end{bmatrix}$$

Example (cont)

$$\begin{bmatrix} 2 & 1 & 0 \\ 1 & 0 & 4 \end{bmatrix} \begin{bmatrix} 3 & 1 & 2 & 1 \\ 1 & 0 & 1 & 2 \\ 5 & 4 & 1 & 1 \end{bmatrix} \begin{bmatrix} 7 & c_{12} & c_{13} & c_{14} \\ c_{21} & c_{22} & c_{23} & c_{24} \end{bmatrix}$$

Example (cont)

$$\begin{bmatrix} 2 & 1 & 0 \\ 1 & 0 & 4 \end{bmatrix} \begin{bmatrix} 3 & 1 & 2 & 1 \\ 1 & 0 & 1 & 2 \\ 5 & 4 & 1 & 1 \end{bmatrix} \begin{bmatrix} 7 & c_{12} & c_{13} & c_{14} \\ c_{21} & c_{22} & c_{23} & c_{24} \end{bmatrix}$$

$$\begin{bmatrix} 2 & 1 & 0 \\ 1 & 0 & 4 \end{bmatrix} \begin{bmatrix} 3 & 1 & 2 & 1 \\ 1 & 0 & 1 & 2 \\ 5 & 4 & 1 & 1 \end{bmatrix} \begin{bmatrix} 7 & 2 & 5 & 4 \\ 23 & 17 & 6 & 5 \end{bmatrix}$$

Practice

Sales data are summaried in the matrix

$$A = \begin{array}{c} \\ \text{group A} \\ \text{group B} \end{array} \begin{array}{c} \begin{array}{ccc} \textit{peanuts} & \textit{soda} & \text{hot dogs} \end{array} \\ \begin{pmatrix} 8 & 5 & 12 \\ 15 & 7 & 13 \end{pmatrix} \end{array}$$

and sell prices is represented in the matrix

$$B = \begin{array}{r} \\ \textit{peanuts} \\ \textit{soda} \\ \text{hot dogs} \end{array} \begin{array}{c} \begin{array}{cccc} \textbf{Store 1} & \textbf{Store 2} & \textbf{Store 3} & \textbf{Store 4} \end{array} \\ \begin{pmatrix} 2 & 2.5 & 2 & 3 \\ 2.7 & 2.6 & 3 & 2.5 \\ 3 & 3 & 2.5 & 2.5 \end{pmatrix} \end{array}$$

Compute the product AB and give an interpretation for each element in the product.

Power of a square matrix

If A is a square matrix then

$$A^k = \underbrace{AA\ldots A}_{k \quad times}$$

and $A^0 = \mathbf{I}_n$

where

Identity matrix

The identity matrix $\mathbf{I}_n$ or simply denoted by $\mathbf{I}$ is the $n \times n$ matrix with 1s on the main diagonal (upper left to lower right), and zeros elsewhere.

$$\mathbf{I}_1 = \begin{bmatrix} 1 \end{bmatrix}, \quad \mathbf{I}_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{I}_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{I}_n = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \\ 0 & 0 & 0 & \cdots & 1 \end{bmatrix}$$

Practice

Compute A^2 and B^3 where

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 2 & 4 \\ 1 & 1 & 1 \end{bmatrix}$$

and

$$B = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 5 \end{bmatrix}$$

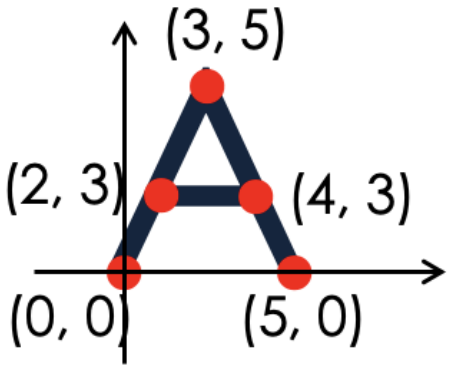
Properties of Multiplication Matrix

1. $A(B+C) = AB + AC$ (distributive law)
2. $(A+B)C = AC + BC$ (distributive law)
3. $A(BC) = (AB)C$ (associate law)
4. $AB = BA$ **not true** (**not communicate**)
5. $A\mathbf{I} = A$
6. $\mathbf{I}B = B$
7. $a(AB) = (aA)B = A(aB)$
8. $(AB)^T = B^T A^T$ (reverse order)

An Application in Computer Graphics

Computer graphics deals with images displayed on a computer screen, and so arises in a variety of applications, ranging from word processors, to Star Wars animations, to video games, to wire-frame images of an airplane. These images consist of a number of points on the screen, together with instructions on how to fill in areas bounded by lines and curves. Often curves are approximated by a set of short straight-line segments, so that the curve is specified by a series of points on the screen at the end of these segments. Matrix transformations are important here because matrix images of straight line segments are again line segments. Note that a colour image requires that three images are sent, one to each of the red,green, and blue phosphorus dots on the screen, in varying intensities.

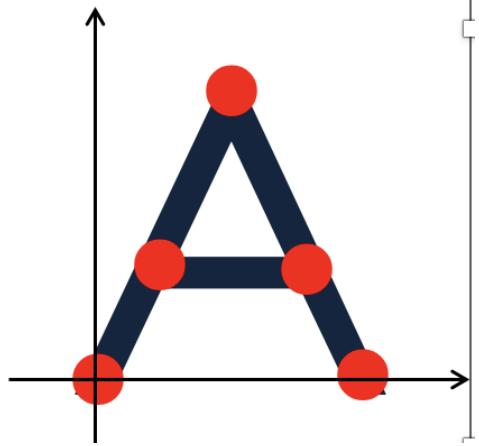
Example - zoom in



$$D=\begin{pmatrix} 0 & 5 & 2 & 4 & 3 \\ 0 & 0 & 3 & 3 & 5 \end{pmatrix}$$

If $A = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$ then the result of AD is

$$\begin{pmatrix} 0 & 10 & 4 & 8 & 6 \\ 0 & 0 & 6 & 6 & 10 \end{pmatrix}$$



Slant an image

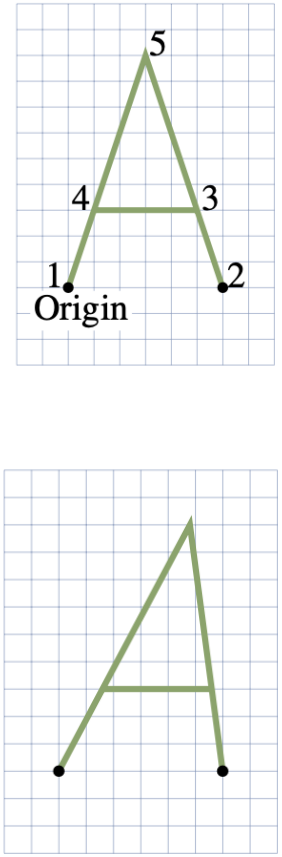
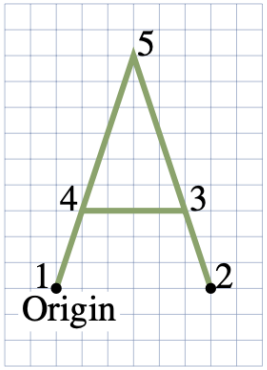


Figure 1: slant the letter to the right by multiplying by an x-shear matrix $A = \begin{bmatrix} 1 & 0.2 \\ 0 & 1 \end{bmatrix}$

Shrink the x-coordinate



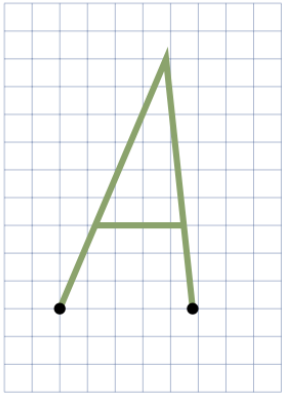


Figure 2: shrink x -coordinate by 0.8 with t an x-scale matrix $B = \begin{bmatrix} 0.8 & 0 \\ 0 & 1 \end{bmatrix}$

Rotate about origin

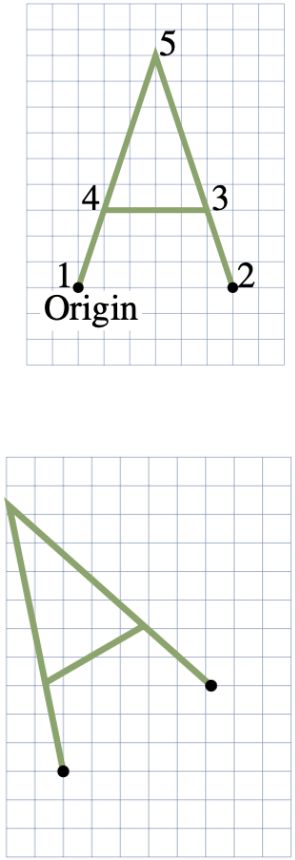


Figure 3: rotate the letter about the origin through 30° by matrix

$$R_{\frac{\pi}{6}} = \begin{bmatrix} \cos(\pi/6) & -\sin(\pi/6) \\ \sin(\pi/6) & \cos(\pi/6) \end{bmatrix}$$

Exercise

What is the result of the letter after applying a transform with matrix

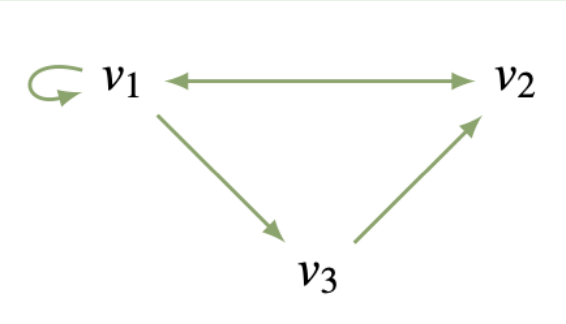
$$B = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

Application to Directed Graph

Definition (Adjacency matrix)

A d**irected graph** consists of a set of points (called **vertices**) connected by arrows (called **edges**). The **adjacency** matrix A of the graph is a square matrix whose (i, j) - entry is 1 if there is an edge from v_j to v_i and zero otherwise

Example



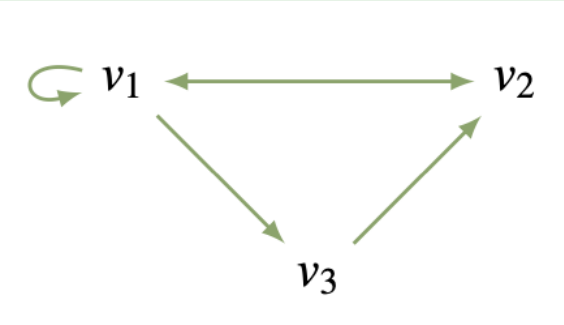
Adjacency matrix

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

Theorem

If A is the adjacency matrix of a directed graph with n vertices, then the (i, j)-entry of A^r is the number of paths of length r (also called by r-paths) from $v_j \to v_i$.

Example



Adjacency matrix $A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}$. Then

$$A^2 = \begin{bmatrix} 2 & 1 & 1 \\ 2 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}, \; A^3 = \begin{bmatrix} 4 & 2 & 1 \\ 3 & 2 & 1 \\ 2 & 1 & 1 \end{bmatrix}$$

Since $(2, 1)$-entry of A^2 is 2, there are two paths of length 2 from v_1 to v_2. Similar no entry of A^3 is zero show that it is possible to go from any vertex to any other vertex in exactly three steps.

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Inverse matrix

- If A is a square matrix, a matrix B is called an inverse of A if and only if

$$AB = \mathbf{I} \text{ and } BA = \mathbf{I}$$

where $\mathbf{I} = \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{bmatrix}$ is an identity matrix of the same size as A

- A matrix A that **has an inverse** is called an **invertible** matrix or **nonsingular**. Otherwise, it is called noninvertible or singular
- If A is an invertible matrix, the inverse of A is denoted A^{-1}
- A^{-1} (when it exists) is a square matrix of the **same size** as A with the property that

$$AA^{-1} = A^{-1}A = \mathbf{I}$$

Example

Show that $B = \begin{bmatrix} -1 & 1 \\ 1 & 0 \end{bmatrix}$ is an inverse of $A = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$.

Example

Show that $B = \begin{bmatrix} -1 & 1 \\ 1 & 0 \end{bmatrix}$ is an inverse of $A = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$.

Solution

We have

$$AB = BA = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \equiv \boldsymbol{I}_2$$

Hence B is an inverse of A

Uniqueness of Inverse If Exists

Suppose that A is invertible and A^{-1} is an inverse of A. If B is also an inverse of A then we have

$$AB = BA = \mathbf{I}$$

From the propery of multiplication by an identity matrix, we have

$$B = B\mathbf{I} = B(AA^{-1}) = (BA)A^{-1} = \mathbf{I}A^{-1} = A^{-1}$$

That is $B = A^{-1}$.

Example - Find the inverse of a matrix

The inverse of

$$A = \begin{bmatrix} 3 & 1 \\ 5 & 2 \end{bmatrix}$$

if exists must be have the same size 2×2 as matrix A. That is it must have the form

$$\begin{bmatrix} x & y \\ z & w \end{bmatrix}$$

which satisfies

$$A \begin{bmatrix} x & z \\ y & w \end{bmatrix} = \begin{bmatrix} x & z \\ y & w \end{bmatrix} A = \mathbf{I}_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Example - Find the inverse of a matrix

The inverse of

$$A = \begin{bmatrix} 3 & 1 \\ 5 & 2 \end{bmatrix}$$

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which satisfies

$$A \begin{bmatrix} x & z \\ y & w \end{bmatrix} = \begin{bmatrix} x & z \\ y & w \end{bmatrix} A = \mathbf{I}_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$A \begin{bmatrix} x & z \\ y & w \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ yields

$$\begin{bmatrix} 3 & 1 \\ 5 & 2 \end{bmatrix} \begin{bmatrix} x & z \\ y & w \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

So we need to solve two linear systems to find values of the entries x, y, z, w

$$\begin{bmatrix} 3 & 1 \\ 5 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

and

$$\begin{bmatrix} 3 & 1 \\ 5 & 2 \end{bmatrix} \begin{bmatrix} z \\ w \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

by Gaussian algorithm to carry the augmented matrices into reduced row echelon form

Reduction procedure for the first system

$$\left[\begin{array}{cc|c} 3 & 1 & 1 \\ 5 & 2 & 0 \end{array}\right] \xrightarrow{\frac{r_1}{3}} \left[\begin{array}{cc|c} 1 & \frac{1}{3} & \frac{1}{3} \\ 5 & 2 & 0 \end{array}\right] \xrightarrow{r_2-5r_1} \left[\begin{array}{cc|c} 1 & \frac{1}{3} & \frac{1}{3} \\ 0 & \frac{1}{3} & -\frac{5}{3} \end{array}\right]$$

$$\xrightarrow{3r_2} \left[\begin{array}{cc|c} 1 & \frac{1}{3} & \frac{1}{3} \\ 0 & 1 & -5 \end{array}\right] \xrightarrow{r_1-\frac{1}{3}r_2} \left[\begin{array}{cc|c} 1 & 0 & \mathbf{2} \\ 0 & 1 & \mathbf{-5} \end{array}\right]$$

$$\Rightarrow \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 \\ -5 \end{bmatrix}$$

and for the second system

$$\left[\begin{array}{cc|c} 3 & 1 & 0 \\ 5 & 2 & 1 \end{array}\right] \xrightarrow{\frac{r_1}{3}} \left[\begin{array}{cc|c} 1 & \frac{1}{3} & 0 \\ 5 & 2 & 1 \end{array}\right] \xrightarrow{r_2-5r_1} \left[\begin{array}{cc|c} 1 & \frac{1}{3} & 0 \\ 0 & \frac{1}{3} & 1 \end{array}\right]$$

$$\xrightarrow{3r_2} \left[\begin{array}{cc|c} 1 & \frac{1}{3} & 0 \\ 0 & 1 & 3 \end{array}\right] \xrightarrow{r_1-\frac{1}{3}r_2} \left[\begin{array}{cc|c} 1 & 0 & \mathbf{-1} \\ 0 & 1 & \mathbf{3} \end{array}\right]$$

$$\Rightarrow \begin{bmatrix} z \\ w \end{bmatrix} = \begin{bmatrix} -1 \\ 3 \end{bmatrix}$$

Hence

$$\begin{bmatrix} x & z \\ y & w \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ -5 & 3 \end{bmatrix}$$

Need to continue to verify that

$$\begin{bmatrix} 2 & -1 \\ -5 & 3 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 5 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

to conclude that

$$A^{-1} = \begin{bmatrix} 2 & -1 \\ -5 & 3 \end{bmatrix}$$

Remark

Both linear systems have the same coefficient matrix A, so exactly the same row operations were used to solve both systems. In each case, the end result was something of the form $[I_2|X]$ where X gave a column of the inverse. So to simplify the procedure, we can combine them into problem of carry the matrix

$$\left[\begin{array}{cc|cc} 3 & 1 & 1 & 0 \\ 5 & 2 & 0 & 1 \end{array}\right]$$

to reduced echelon form

$$\left[\begin{array}{cc|cc} 1 & 0 & 2 & -1 \\ 0 & 1 & -5 & 3 \end{array}\right]$$

with the same sequence of elementary row operations as before and read the inverse from the half on the right of the final matrix. This exploration motivates the following algorithm to find an inverse of a matrix if it exists

The Gauss − Jordan Method to Find Matrix Inverse

If A is an invertible (square) matrix, there exists a sequence of elementary row operations that carry A to the identity matrix I of the same size, written $A \to \mathbf{I}$. This same series of row operations carries $\mathbf{I}$ to A^{-1}; that is, $\mathbf{I} \to A^{-1}$. The algorithm can be summarized as follows:

$$\begin{bmatrix} A & \mathbf{I} \end{bmatrix} \to \begin{bmatrix} \mathbf{I} & A^{-1} \end{bmatrix}$$

This procedure will be justified in section Elementary Matrices

Example

Find inverse of

$$\begin{bmatrix} \frac{1}{2} & 1 \\ 1 & 3 \end{bmatrix}$$

Solution

- **Step 1** Set up the augmented matrix

$$\left[\begin{array}{cc|cc} \frac{1}{2} & 1 & 1 & 0 \\ 1 & 3 & 0 & 1 \end{array}\right]$$

$$\underbrace{\qquad\qquad}_{A} \qquad \underbrace{\qquad\qquad}_{I_2}$$

Solution

- **Step 1** Set up the augmented matrix

$$\left[\begin{array}{cc|cc} \frac{1}{2} & 1 & 1 & 0 \\ 1 & 3 & 0 & 1 \end{array}\right]$$
$$\underbrace{\qquad\qquad}_{A} \qquad \underbrace{\qquad\qquad}_{I_2}$$

- **Step 2** Use Gaussian algorithm to carry the augmented matrix to reduced row echelon form

$$\left[\begin{array}{cc|cc} \frac{1}{2} & 1 & 1 & 0 \\ 1 & 3 & 0 & 1 \end{array}\right] \xrightarrow[R_2+(-1)R_1]{2R_1} \left[\begin{array}{cc|cc} 1 & 2 & 2 & 0 \\ 0 & 1 & -2 & 1 \end{array}\right] \xrightarrow{R_1+(-2)R_2} \left[\begin{array}{cc|cc} 1 & 0 & 6 & -2 \\ 0 & 1 & -2 & 1 \end{array}\right].$$

Solution

- **Step 1** Set up the augmented matrix

$$\left[\begin{array}{cc|cc} \frac{1}{2} & 1 & 1 & 0 \\ 1 & 3 & 0 & 1 \end{array}\right]$$
$$\underbrace{\qquad\qquad}_{A} \qquad \underbrace{\qquad\qquad}_{I_2}$$

- **Step 2** Use Gaussian algorithm to carry the augmented matrix to reduced row echelon form

$$\left[\begin{array}{cc|cc} \frac{1}{2} & 1 & 1 & 0 \\ 1 & 3 & 0 & 1 \end{array}\right] \xrightarrow[R_2+(-1)R_1]{2R_1} \left[\begin{array}{cc|cc} 1 & 2 & 2 & 0 \\ 0 & 1 & -2 & 1 \end{array}\right] \xrightarrow{R_1+(-2)R_2} \left[\begin{array}{cc|cc} 1 & 0 & 6 & -2 \\ 0 & 1 & -2 & 1 \end{array}\right].$$

- **Step 3** Read the inverse on the right hand side

$$A^{-1} = \begin{bmatrix} 6 & -2 \\ -2 & 1 \end{bmatrix}$$

Practice

Find the inverse of the matrices

$$A = \begin{bmatrix} 2 & 7 & 1 \\ 1 & 4 & -1 \\ 1 & 3 & 0 \end{bmatrix}$$

and

$$B = \begin{bmatrix} 1 & 2 & 2 \\ 1 & 0 & 2 \\ 2 & 2 & 4 \end{bmatrix}$$

if exist.

Example (Cancellation rule)

Assume that A is invertible

1. If $AB = AC$ then $B = C$
2. If $BA = CA$ then $B = C$

Example (Cancellation rule)

Assume that A is invertible

1. If $AB = AC$ then $B = C$
2. If $BA = CA$ then $B = C$

Solution

1. *We have*

$$B = IB = (A^{-1}A)B = A^{-1}(AB) = A^{-1}(AC) = (A^{-1}A)C = IC = C$$

Hence $B = C$

2. *Similar*

$$B = BI = B(AA^{-1}) = (BA)A^{-1} = (CA)A^{-1} = C(AA^{-1}) = CI = C$$

Hence $B = C$

Question

If A is not invertible, we we conclude that $B = C$ if $AB = AC$. If not, please give a counter example for which $AB = AC$ but $B \neq C$

Use matrix inverse to solve system of linear equations

- A system of linear equation can be written is a single matrix equation

$$A\mathbf{x} = b$$

where A, b are known and $\mathbf{x}$ is to be determined.

- If A **is invertible** then multiply each side of the equation on the left by A^{-1} to get

$$\begin{aligned} A^{-1}A\mathbf{x} &= A^{-1}b \\ \mathbf{I}\mathbf{x} &= A^{-1}b \\ \mathbf{x} &= A^{-1}b \end{aligned}$$

Example

Solve the system equations

$$3x_1 + x_2 = 3$$
$$5x_1 + 2x_2 = 8$$

Solution

Matrix form of the system is $A\boldsymbol{x} = \boldsymbol{b}$ *where* $A = \begin{bmatrix} 3 & 1 \\ 5 & 2 \end{bmatrix}$, $\boldsymbol{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$, $\boldsymbol{b} = \begin{bmatrix} 3 \\ 8 \end{bmatrix}$.

A *has the inverse* $A^{-1} = \begin{bmatrix} 2 & -1 \\ -5 & 3 \end{bmatrix}$. *So the solution of the system is*

$$\boldsymbol{x} = A^{-1}\boldsymbol{b} = \begin{bmatrix} 2 & -1 \\ -5 & 3 \end{bmatrix} \begin{bmatrix} 3 \\ 8 \end{bmatrix} = \begin{bmatrix} -2 \\ 9 \end{bmatrix} \text{ or } (x_1 = -2, x_2 = 9)$$

Practice

Use the inverse of a suitable matrix to give the solutions to the system

$$\begin{cases} x \qquad\quad + z = 1 \\ x - y + z = 3 \\ x + y - z = 2 \end{cases}$$

Properties of Inverse

If A and B are invertible matrices of size $n \times n$ then

1. I is invertible and $I^{-1} = I$.
2. A^{-1} is invertible , and $(A^{-1})^{-1} = A$
3. **Inverse of product** AB is invertible and $(AB)^{-1} = B^{-1}A^{-1}$ (**reverse order**).
4. A^k is invertible for any $k \geq 1$, and **inverse of power** $(A^k)^{-1} = (A^{-1})^k$ (**power of inverse**).
5. aA is invertible for $a \neq 0$ and $(aA)^{-1} = \frac{1}{a}A^{-1}$.
6. **Inverse of transpose** $(A^T)^{-1} = (A^{-1})^T$.

Example

Find A if $(A^T - 2I)^{-1} = \begin{bmatrix} 2 & 1 \\ -1 & 0 \end{bmatrix}$

Example

Find A if $(A^T - 2I)^{-1} = \begin{bmatrix} 2 & 1 \\ -1 & 0 \end{bmatrix}$

Solution

We have

$$A^T - 2I = \left[(A^T - 2I)^{-1}\right]^{-1} = \begin{bmatrix} 2 & 1 \\ -1 & 0 \end{bmatrix}^{-1} = \begin{bmatrix} 0 & -1 \\ 1 & 2 \end{bmatrix}$$

Hence

$$A^T = 2I + \begin{bmatrix} 0 & -1 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ 1 & 4 \end{bmatrix}$$

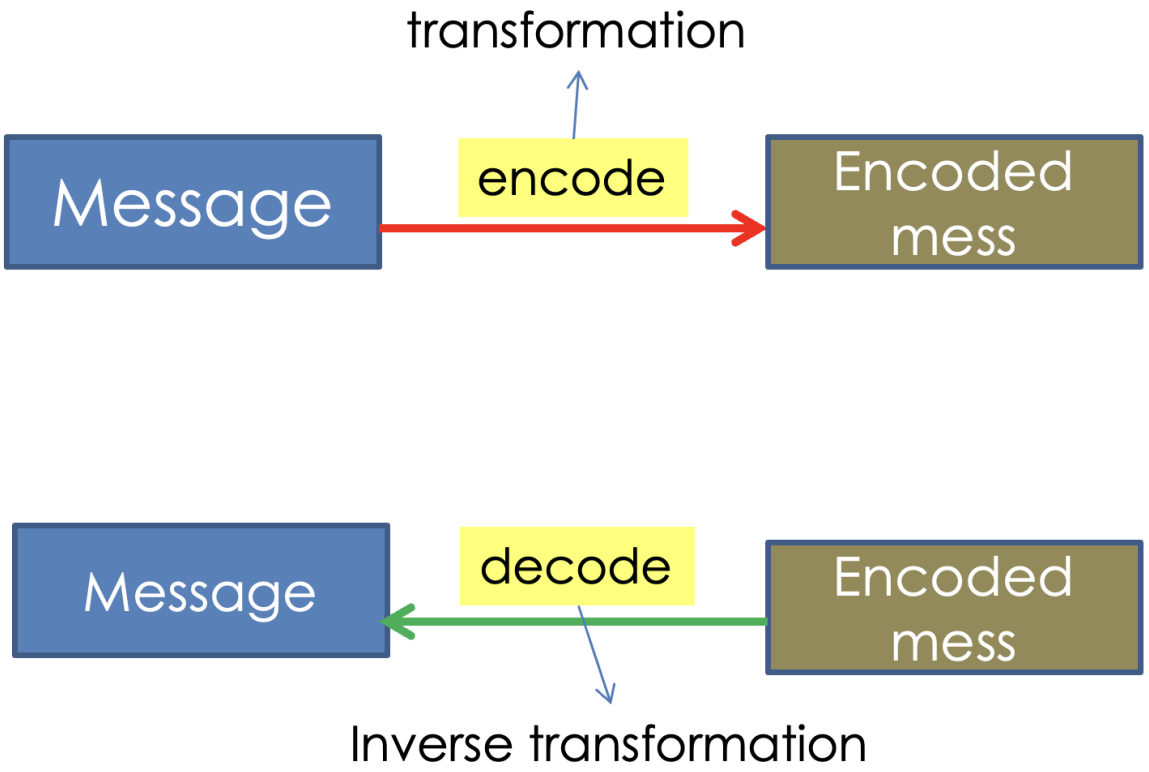
So $A = (A^T)^T = \begin{bmatrix} 2 & 1 \\ -1 & 4 \end{bmatrix}$

Conditions for Existence of Inverse Matrix

The following conditions are equivalent for an $n \times n$ matrix A

1. A is invertible.
2. The homogeneous system $A\mathbf{x} = \mathbf{0}$ has only the trivial solution $\mathbf{x} = \mathbf{0}$.
3. A can be carried to the identity matrix $\mathbf{I}_n$ by elementary row operations.
4. The system $A\mathbf{x} = \mathbf{b}$ has at least one solution $\mathbf{x}$ for every choice of column $\mathbf{b}$.
5. There exists an $n \times n$ matrix C such that $AC = \mathbf{I}_n$.

Cryptography



Example of Encryption a Letter

0 = __	14 = N
1 = A	15 = O
2 = B	16 = P
3 = C	17 = Q
4 = D	18 = R
5 = E	19 = S
6 = F	20 = T
7 = G	21 = U
8 = H	22 = V
9 = I	23 = W
10 = J	24 = X
11 = K	25 = Y
12 = L	26 = Z
13 = M	

Cryption with matrix multiplication and modula

Encrypt a message of 4 letters such as STOP as following

- Represent message $\underbrace{S}_{19}\underbrace{T}_{20}\underbrace{O}_{15}\underbrace{P}_{16}$ by matrix $D = \begin{bmatrix} 19 & 20 \\ 15 & 16 \end{bmatrix}$
- We use an **encoding matrix** $M = \begin{bmatrix} 1 & -3 \\ -1 & 4 \end{bmatrix}$ to encrypt the message by transformation

$$MD = \begin{bmatrix} 1 & -3 \\ -1 & 4 \end{bmatrix} \begin{bmatrix} 19 & 20 \\ 15 & 16 \end{bmatrix} = \begin{bmatrix} -26 & -28 \\ 41 & 44 \end{bmatrix}$$

- The encoded message is $\underbrace{-26}_{A}\underbrace{-28}_{Z}\underbrace{41}_{N}\underbrace{44}_{Q}$

  if we consider a modula representation for a letter, e.g, n presents letter A if and only if $n \mod 27 = 1$

Encryption with inverse matrix

You receive an encoded message. Need to encrypt to find the origin message.

Encryption with inverse matrix

You receive an encoded message. Need to encrypt to find the origin message.

The **principal** is as following: If we denote D and E be the matrix representation for origin message and encoded message then

$$E = MD$$

Given encoded matrix E and transformation matrix M, the decoded matrix D is hence obtained with **decoding matrix** M^{-1} as following

$$D = M^{-1}E$$

Then you can find what origin message is. **Maybe**

Example

Want to decode the message

$$E = \begin{bmatrix} 21 & 37 \\ 26 & 53 \end{bmatrix}$$

that was encoded using matrix $M = \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix}$.

Example

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$$E = \begin{bmatrix} 21 & 37 \\ 26 & 53 \end{bmatrix}$$

that was encoded using matrix $M = \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix}$.

Let D be a matrix representing the origin message then

$$E = MD \Rightarrow D = M^{-1}E \text{ with } M^{-1} = \begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix}$$

So $E = \begin{bmatrix} 11 & 5 \\ 5 & 16 \end{bmatrix}$ stands for the message KEEP

The Hill Cipher

1. Codes can use finite fields as alphabets: letters in the message become numbers 0, 1, $\ldots$, $p-1$
2. The numbers are added and multiplied $\mod p$. Divide by p, keep the remainder
3. A Hill Cipher multiplies blocks of the message by a secrete matrix $E(\mod p)$
4. To decode, multiply each block by the inverse matrix $D(\mod p)$. Not a very secure cipher

Section 10.7 in the textbook of Strang, Section 8.8 in the textbook of Nicholson

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Elementary Matrices

Elementary row operations takes an important role in linear algebra: solving linear system (using gaussian algorithm), inverting a matrix (using Gauss-Jordan method). It turns out that they can be performed by left multiplying by certain invertible matrices

Definition

An $n \times n$ matrix E is called an **elementary matrix** if it can be obtained from the identity matrix In by a single elementary row operation (called the operation corresponding to E). Type of E is corresponding to type of row operation:

- **Interchange two rows**: E is of type I
- **Multiply one row by a nonzero number**: E is of type II
- **Add a multiple of one row to a different row**: E is of type III

Example

$$E_1 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$: interchanging rows 1 and 2: elementary of type I

$$E_2 = \begin{bmatrix} 1 & 0 \\ 0 & 9 \end{bmatrix}$$: multiplying row 2 by 9: elementary of type II

$$E_3 = \begin{bmatrix} 1 & 5 \\ 0 & 1 \end{bmatrix}$$: adding 5 times row 2 to row 1: elementary of type III

More concretely, consider $A = \begin{bmatrix} a & b & c \\ p & q & r \end{bmatrix}$ then

$$E_1 A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} a & b & c \\ p & q & r \end{bmatrix} = \begin{bmatrix} p & q & r \\ a & b & c \end{bmatrix}$$ interchange row 1 and row 2

Try to compute $E_2 A$ and $E_3 A$ to see that the results are the same as we apply the corresponding elementary operation for the identity matrix to obtain E_2 and E_3

Elementary Matrices vs Elementary Row Operations

1. If an elementary row operation is performed on an $m \times n$ matrix A, the result is EA where E is the elementary matrix obtained by performing the same operation on the $m \times m$ identity matrix.
2. Every elementary matrix E is invertible, and E^{-1} is also a elementary matrix (of the same type). Moreover, E^{-1} corresponds to the inverse of the row operation that produces E.

The following table gives the inverse of each type of elementary row operation:

Type	**Operation**	**Inverse Operation**
I	Interchange rows p and q	Interchange rows p and q
II	Multiply row p by $k \neq 0$	Multiply row p by $1/k$
III	Add k times row p to row $q \neq p$	Subtract k times row p from row q

Example

$$E_1 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

corresponds to exchanging rows 1 and 2 (type I). So the inverse

$$E_1^{-1} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Example

$$E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 9 \end{bmatrix}$$

corresponds to multiplyring row 3 by 9 (type II). So the inverse

$$E_2^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \frac{1}{9} \end{bmatrix}$$

Example

$$E_3 = \begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

corresponds to adding 5 times row 3 to row 1 (type III). So the inverse

$$E_3^{-1} = \begin{bmatrix} 1 & 0 & -5 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

is to substract 5 times row 3 from row 1

Inverse and Elementary Matrices

Suppose that a matrix A is carried to matrix B, denoted by$A \to B$ by a series of k elementary row operations corresponding to the elementary matrices $E_1, \ldots, E_k$ then the reduction becomes

$$A \to E_1 A \to E_2 E_1 A \to \cdots \to E_k \ldots E_2 E_1 A = B$$

In other word

$$A \to UB \text{ where } U = E_k \ldots E_2 E_1 \text{ is invertible}$$

U can be computed by

$$\begin{bmatrix} A & I \end{bmatrix} \to \begin{bmatrix} B & U \end{bmatrix} \text{ using the operations carrying } A \to B$$

If A is invertible then A can be carried to I, that is

$$\begin{bmatrix} A & I \end{bmatrix} \to \begin{bmatrix} I & U \end{bmatrix} \text{ where } UA = I$$

Hence $U = A^{-1}$. So the Gauss - Jordan algorithm is valid

Furthermore

$$A^{-1} = U \Rightarrow A = U^{-1} = E_1^{-1} E_2^{-1} \dots E_k^{-1}$$

Theorem

A square matrix is invertible if and only if it is a product of elementary matrices.

Example

Express $A = \begin{bmatrix} -2 & 3 \\ 1 & 0 \end{bmatrix}$ as a product of elementary matrices

Example

Express $A = \begin{bmatrix} -2 & 3 \\ 1 & 0 \end{bmatrix}$ as a product of elementary matrices

Solution

The reduction of $A \to I$ is as follow

$$A = \begin{bmatrix} -2 & 3 \\ 1 & 0 \end{bmatrix} \xrightarrow[E_1]{r_1 \leftrightarrow r_2} A = \begin{bmatrix} 1 & 0 \\ -2 & 3 \end{bmatrix} \xrightarrow[E_2]{r_2 + 2r_1} \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix} \xrightarrow[E_3]{\frac{1}{3} r_2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

where $E_1 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, $E_2 = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$, $E_3 = \begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{3} \end{bmatrix}$

Hence $E_3 E_2 E_1 A = I$, *so*

$$A = E_1^{-1} E_2^{-1} E_3^{-1} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}$$

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Triangular Matrices

For a matrix $A = \left[a_{ij}\right]$, the elements $a_{11}, a_{22}, \ldots,$ form the **main diagonal** of A.

- A is called **upper triangular** if every entry below and to the left of the main diagonal is 0
- A is called **lower triangular** if every entry above and to the right of the main diagonal is 0

Upper Triangular Matrix

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ 0 & a_{22} & a_{23} & \cdots & a_{2n} \\ 0 & 0 & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \cdots & a_{nn} \end{bmatrix}$$

Lower Triangular Matrix

$$\begin{bmatrix} a_{11} & 0 & 0 & \cdots & 0 \\ a_{21} & a_{22} & 0 & \cdots & 0 \\ a_{31} & a_{32} & a_{33} & \cdots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \cdots & a_{nn} \end{bmatrix}$$

Example

Solve the system

$$\begin{cases} x_1 + 2x_2 - 3x_3 - x_4 + 5x_5 = 3 \\ \qquad\qquad\quad 5x_3 + x_4 + \ \ x_5 = 8 \\ \qquad\qquad\qquad\qquad\qquad 2x_5 = 6 \end{cases}$$

where the coefficient matrix is upper triangular

It is easy to use **backward substituition** to find the solution of the system equation. Similary, if the coefficient matrix is lower triangular then the system can be solve by **forward substitution** where earlier variables are substituted into later equation

Solve $A\mathbf{x} = \mathbf{b}$ if A is LU-factorizable

If $A = LU$ where L is lower triangular and U is upper triangular the the system $A\mathbf{x} = \mathbf{b}$ can be solved in two stages as following

1. Solve $A\mathbf{y} = \mathbf{b}$ for $\mathbf{y}$ by forward substitution
2. Solve $U\mathbf{x} = \mathbf{y}$ for $\mathbf{x}$ by backward substitution

Matrix A can be carried to a row-echelon matrix U (that is upper triangular) with reduction

$$A \to E_1 A \to E_2 E_1 A \to \cdots \to E_k E_{k-1} \dots E_2 E_1 A = U$$

where $E_1, \dots, E_k$ are elementary matrices corresponding to the row operation used for zero out entries below the leadings entry. Hence

$$A = LU$$

where $L = (E_k E_{k-1} \dots E_2 E_1)^{-1} = E_1^{-1} E_2^{-1} \dots E_k^{-1}$.
If all E_i is lower triangluar and so is E_i^{-1}, we have L is also lower triangular

LU-factorization

If A can be lower reduced to a row - echelon matrix U then

$$A = LU$$

where L is lower triangular and U is upper triangular. The factorization $A = LU$ is called the **LU-factorization** of A

There are many methods for obtaining an LU-factorization of a matrix and LU- factorization is not unique. We show how to obtain an LU-factorization of a matrix by **modifying the Gaussian elimination procedure** in the following example. No row interchanges will be permitted, and we do *not require that the leading entries have value 1*. This is the scheme for **storage of multipliers**.

Example

Find LU-factorization of

$$A = \begin{bmatrix} 6 & -2 & -4 & 4 \\ 3 & -3 & -6 & 1 \\ -12 & 8 & 21 & -8 \\ -6 & 0 & -10 & 7 \end{bmatrix}$$

Solution

We proceed to "zero out" entries below the leading entries, using only the row operation that adds a multiple of one row to a different row.

$$A \xrightarrow[\text{"zero out" the first leading}]{r_2-\frac{1}{2}r_1, r_3+2r_1, r_4+r_1} U_1 \equiv \begin{bmatrix} 6 & -2 & -4 & 4 \\ 0 & -2 & -4 & -1 \\ 0 & 4 & 13 & 0 \\ 0 & -2 & -14 & 11 \end{bmatrix}$$

Build a lower triangular matrix L_1 with 1's on the main diagonal, to record the row operations. Enter the negatives of the multipliers used in the row operations in the first column of L_1, below the first diagonal entry of L_1

$$L_1 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ \frac{1}{2} & 1 & 0 & 0 \\ -2 & * & 1 & 0 \\ -1 & * & * & 1 \end{bmatrix}$$

"Zero out" the second leading of U_1

$$U_1 \xrightarrow[r_3+2r_2, r_4-r_2]{\text{"Zero out" the second leading}} U_2 \equiv \begin{bmatrix} 6 & -2 & -4 & 4 \\ 0 & -2 & -4 & -1 \\ 0 & 0 & 5 & -2 \\ 0 & 0 & -10 & 12 \end{bmatrix}$$

Enter the negatives of the multipliers from the row operations below the second diagonal of L_1 . Call the new matrix L_2.

$$L_2 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ \frac{1}{2} & 1 & 0 & 0 \\ -2 & -2 & 1 & 0 \\ -1 & 1 & * & 1 \end{bmatrix}$$

"Zero out" the second leading of U_2

$$U_2 \xrightarrow[r_4+2r_3]{\text{"Zero out" the second leading}} U_3 \equiv \begin{bmatrix} 6 & -2 & -4 & 4 \\ 0 & -2 & -4 & -1 \\ 0 & 0 & 5 & -2 \\ 0 & 0 & 0 & 8 \end{bmatrix}$$

Enter the negatives of the multipliers from the row operations below the third diagonal of L_2 . Call the new matrix L_3.

$$L_3 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ \frac{1}{2} & 1 & 0 & 0 \\ -2 & -2 & 1 & 0 \\ -1 & 1 & -2 & 1 \end{bmatrix}$$

Then $L = L_3$ and $U = U_3$ and $A = LU$

Practice

Find an LU-factorization for $A = \begin{bmatrix} 2 & 4 & 2 \\ 1 & 1 & 2 \\ -1 & 0 & 2 \end{bmatrix}$